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A Prompts

`cyclic_shift` ($N = 8$, $A = 4$, $k = 4$).

```
You are a deterministic sequence transformer.
Task: Cyclic character shift.
Alphabet: A, B, C, D
Rule: Replace every character with the next one in the alphabet.
Mapping: D -> A, and A -> B.
Constraint: The input contains vertical bars (|). These are structural
boundaries. DO NOT shift them. Copy them exactly to the output.
No spaces, no explanations.

Input: CABA|CCDA
Output:                 (expected: DBCB|DDAB)
```

Spaced and digit formats. The spaced version of this cell shows the input as `C A B A | C C D A` and replaces "No spaces, no explanations." with "Write the output in the same format as the input: every character separated by a single space. No explanations."; everything else is unchanged. The digit version relabels A, B, C, D as $0, 1, 2, 3$ (input `2010|2230`), with the task line, rule, and mapping worded for digits ("Cyclic digit shift", "Replace every digit with the next one in the alphabet"). The same seed draws the same positions in all three formats.

`running_xor`, original and reset ($N = 8$, $k = 4$). Both prompts state the rule $y_i = x_i \oplus x_{i-1}$ with the first digit copied. The reset prompt adds: *"Immediately after each boundary, treat that position exactly like the first digit of the whole input—copy it as-is (nothing before a boundary counts as 'previous'), then resume the XOR rule from there until the next boundary,"* with the worked example `1011|0010` → `1110|0011`. For input `1101|1111`, the original expects `1011|0000`, while the reset variant expects `1011|1000`: only the digit after the marker differs.

`sliding_window_max` ($N = 8$, $k = 4$). The prompt states the rule with a worked example and adds: *"DO NOT include them [the bars] in the window or count them as digits."* For input `5170|1247` the expected output is `5577|7247`.

B Trials per cell

Table 3 gives trials per cell (one cell = task, model, length, density, and alphabet or modulus value) for the data pooled in Figures 4, 1, and 3. Gemini uses 3.

Table 3: Trials per cell, $N \leq 64$. Q2.5: Qwen2.5-Instruct; Q3.8: Qwen3.8-27B; Gemma 4: Gemma-4-31B-it.

Task	Q2.5-7B	Q2.5-14B	Q2.5-32B	Mistral	Phi	Q3.8	Gemma 4
`cyclic_shift`	20	20	20	40	20	20	20
`random_permutation`	40	40	40	40	20	20	20
`conditional_shift`	40	20	40	40	20	20	20
`copy_previous`	40	40	80	40	40	40	40
`running_xor`	40	40	40	40	20	40	40
`running_sum_local`	40	40	80	40	20	40	40
`sliding_window_max`	20	20	40	20	20	20	20
Reset variants	80	–	20	–	–	20	20
Spaced versions (each)	20	20	20	20	20	20	20
Digit versions (each)	20	20	20	20	20	20	20

C Per-model results

Figures 6–8 show accuracy by length for each model (contiguous letters and dependency tasks).

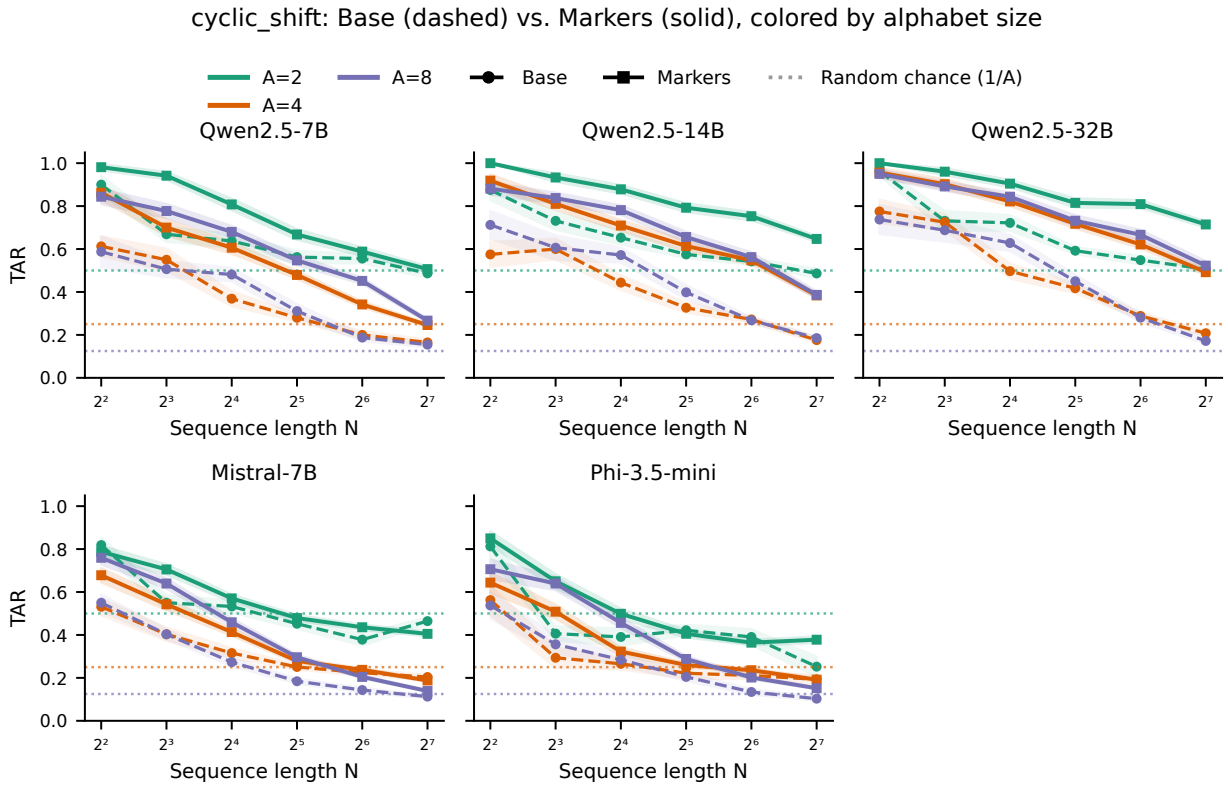


Figure 6: `cyclic_shift` by model and length. Dashed: Base; solid: Bounds (pooled over densities); colors: alphabet size; dotted: chance.

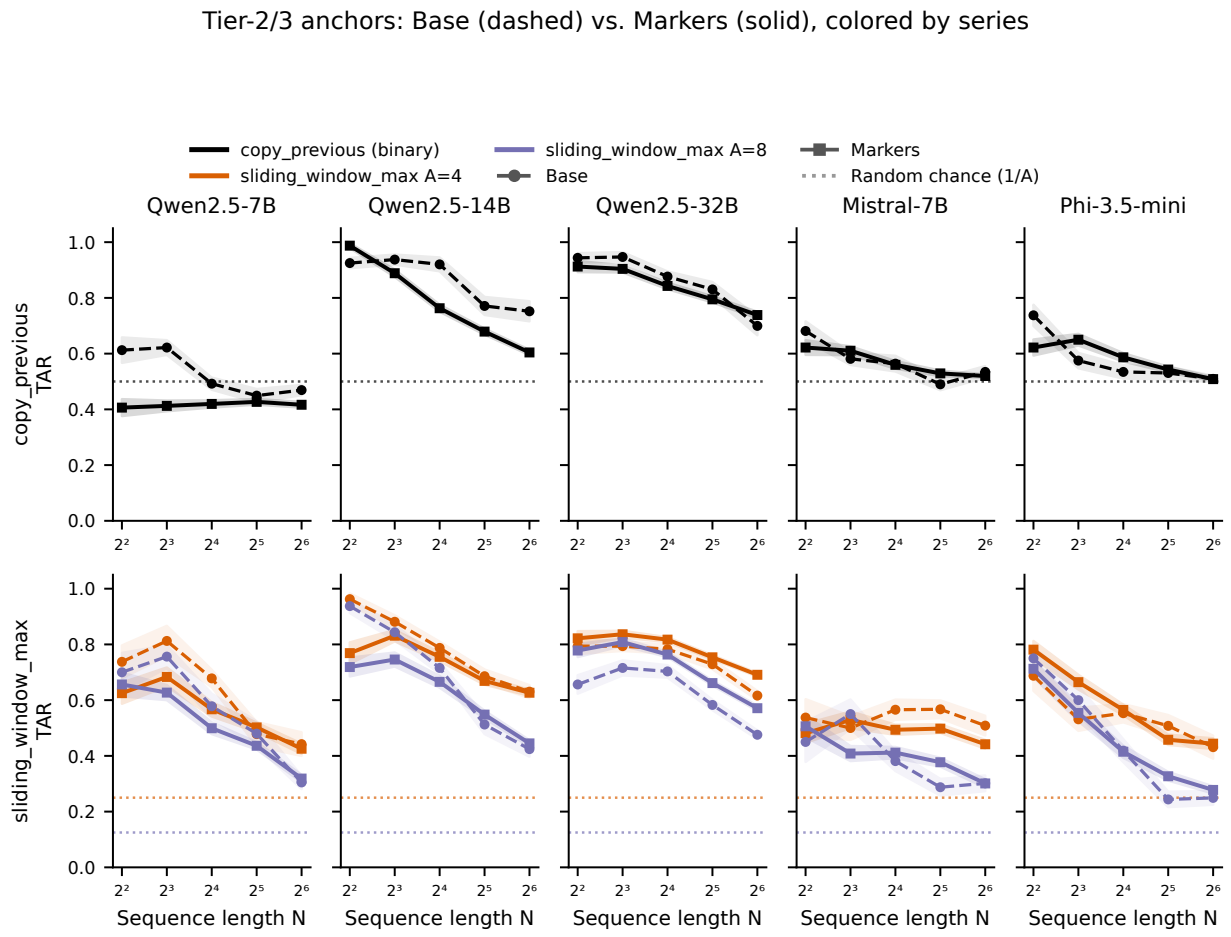


Figure 7: `copy_previous` and `sliding_window_max` by model. `copy_previous` is hurt at every model shown; `sliding_window_max` is hurt at Qwen2.5-7B ($z = 4.6$), 14B ($z = 5.3$), Mistral, and Phi, and helped at Qwen2.5-32B ($z = 5.1$).

Mistral-7B. In all three position-local tasks, the benefit is largest at short lengths and decays to zero at the longest length tested (`cyclic_shift`, $N = 256$: $+0.008$, $z = 0.4$; `random_permutation`, $N = 64$: -0.009, $z = 0.5$; `conditional_shift`, $N = 64$: -0.006, $z = 0.4$). In Qwen models the benefit holds or grows with length.

Repetition collapse. On `running_xor`, Qwen2.5-7B repeats a single wrong digit in about 38% of trials, more often under dense markers. This failure falls to 0.9% at 14B and 0.1% for Mistral-7B, so it is specific to Qwen2.5-7B.

Out-of-alphabet answers. On the digit tasks, Qwen2.5-7B often writes a letter (usually `t`, sometimes `o`) in place of `0`, in 34% to 39% of outputs, with or without markers. On spaced letters, several models step outside the alphabet (for example `I` after `H` instead of wrapping to `A`). Such characters are scored as errors.

D Per-position accuracy

Table 4 gives the per-position means plotted in Figure 2, and Table 5 breaks them down by distance from the marker. "No markers" is the no-marker run's accuracy at the same position indices. Means are over the tasks in each block; per-task values are in `results_clean/_summary/per_position_values.csv`.

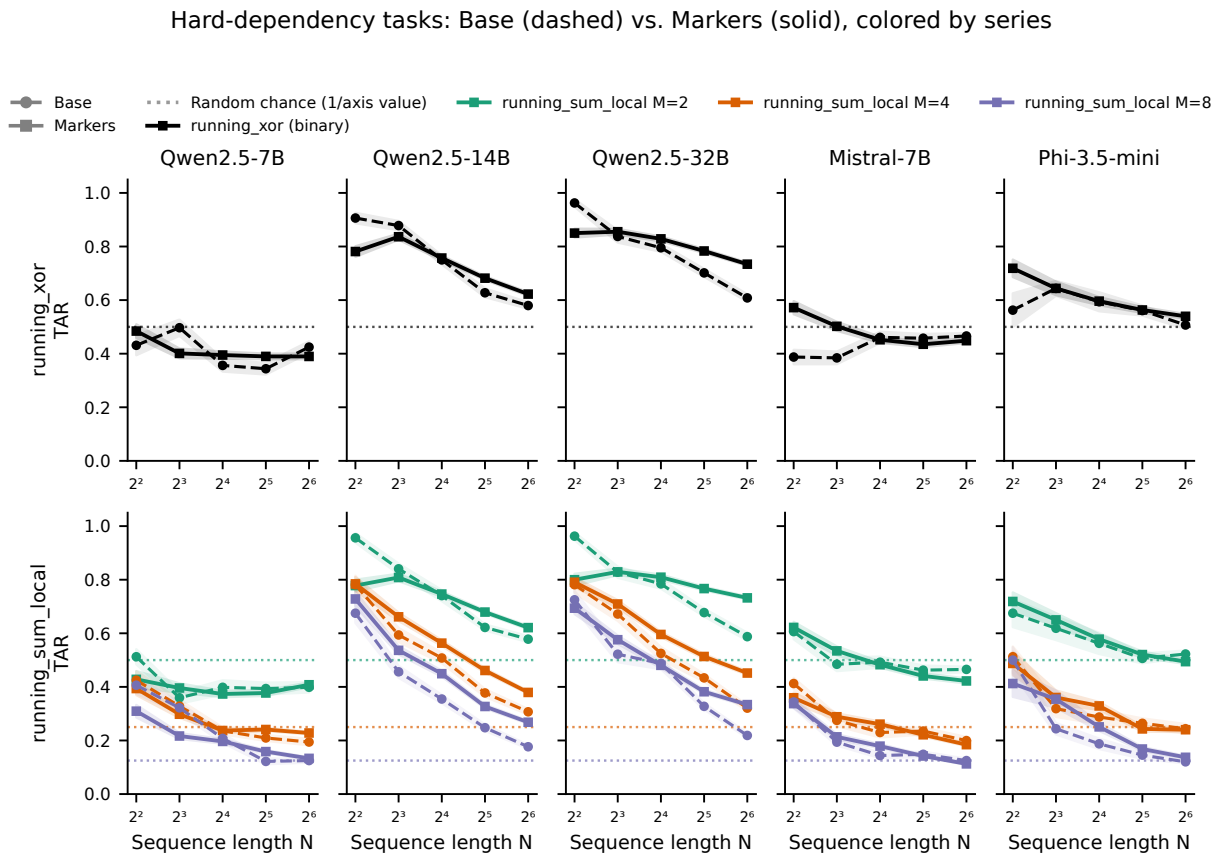


Figure 8: `running_xor` and `running_sum_local` by model.

E Reset variants, pooled

F External tasks

On Kadane's algorithm, base accuracy at both Qwen scales is statistically indistinguishable from chance under a shuffle test, so we do not interpret its effects. Minimum is well above chance at Qwen2.5-32B ($z = 4.1$ from shuffled chance).

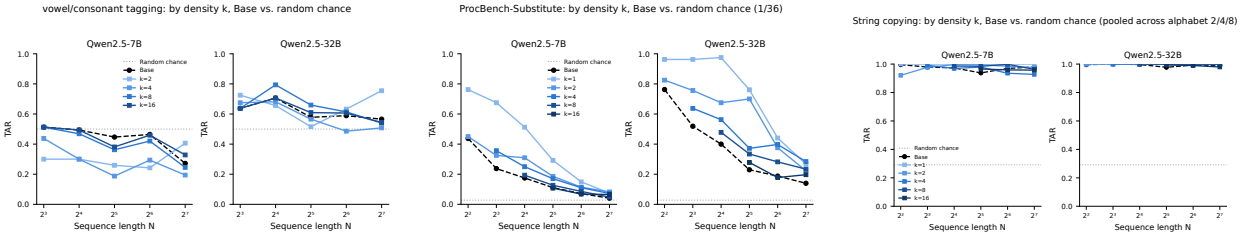


Figure 9: External position-local tasks by density. Left: vowel/consonant tagging. Middle: ProcBench-Substitute. Right: string copying.

Table 4: Per-position accuracy by model ($N \leq 64$, $k \in \{2, 4, 8, 16\}$), mean over the tasks in each block.

	Within a segment			Right after a marker		
Model	No markers	Markers	Δ	No markers	Markers	Δ
Position-local tasks, letters						
Qwen2.5 7B	0.38	0.45	+0.07	0.37	0.51	+0.14
Qwen2.5 14B	0.43	0.55	+0.11	0.40	0.63	+0.23
Qwen2.5 32B	0.49	0.65	+0.16	0.45	0.79	+0.34
Mistral 7B	0.35	0.35	0.00	0.30	0.39	+0.09
Phi-3.5 mini	0.31	0.31	0.00	0.28	0.35	+0.07
Qwen3.8 27B	0.50	0.69	+0.20	0.47	0.81	+0.34
Gemma 4 31B	0.56	0.72	+0.16	0.51	0.80	+0.29
Gemini 3.5 Flash	0.57	0.74	+0.17	0.52	0.83	+0.32
Position-local tasks, spaced letters						
Qwen2.5 7B	0.54	0.55	+0.01	0.52	0.52	0.00
Qwen2.5 14B	0.69	0.80	+0.11	0.67	0.82	+0.15
Qwen2.5 32B	0.79	0.93	+0.14	0.79	0.92	+0.14
Mistral 7B	0.40	0.43	+0.03	0.35	0.45	+0.09
Phi-3.5 mini	0.35	0.35	−0.01	0.33	0.34	+0.01
Qwen3.8 27B	0.83	0.92	+0.09	0.84	0.92	+0.08
Gemma 4 31B	0.88	0.94	+0.06	0.89	0.94	+0.05
Position-local tasks, digits						
Qwen2.5 7B	0.35	0.38	+0.04	0.30	0.35	+0.05
Qwen2.5 14B	0.77	0.85	+0.08	0.77	0.87	+0.09
Qwen2.5 32B	0.83	0.94	+0.11	0.83	0.93	+0.09
Mistral 7B	0.43	0.38	−0.04	0.41	0.39	−0.02
Phi-3.5 mini	0.48	0.55	+0.07	0.44	0.56	+0.12
Qwen3.8 27B	0.85	0.94	+0.09	0.85	0.94	+0.09
Gemma 4 31B	0.92	0.99	+0.07	0.92	0.98	+0.06
Dependency tasks						
Qwen2.5 7B	0.43	0.41	−0.02	0.42	0.36	−0.06
Qwen2.5 14B	0.68	0.68	0.00	0.64	0.49	−0.15
Qwen2.5 32B	0.70	0.77	+0.07	0.66	0.52	−0.15
Mistral 7B	0.44	0.43	0.00	0.43	0.38	−0.04
Phi-3.5 mini	0.49	0.48	0.00	0.45	0.42	−0.03
Qwen3.8 27B	0.68	0.70	+0.02	0.63	0.44	−0.19
Gemma 4 31B	0.79	0.85	+0.05	0.78	0.52	−0.26
Gemini 3.5 Flash	0.92	0.97	+0.05	0.92	0.49	−0.43
Dependency tasks, reset variants						
Qwen2.5 7B	0.43	0.43	0.00	0.41	0.58	+0.17
Qwen2.5 32B	0.69	0.79	+0.10	0.65	0.92	+0.27
Qwen3.8 27B	0.66	0.68	+0.03	0.63	0.98	+0.36
Gemma 4 31B	0.77	0.85	+0.08	0.75	0.97	+0.22

G Mechanism details

This appendix collects the attention analyses behind Section 7: the mechanism overview (Figure 11), attention statistics by layer (Figure 12), example heads (Figure 13), the earlier head ablation (Figure 14), a digit-tokenization check of the attention pattern (Figure 15), error recovery after markers (Figure 16), the head taxonomy (Table 10), and the 7B ablation (Table 9). In Figure 16, the 50% rate away from markers is chance for this binary alphabet.

Table 5: Accuracy by distance after the marker at $k = 16$ ($N \in \{32, 64\}$): 0 is the position right after the marker. Mean over the tasks in each block.

		Positions after the marker				
Tasks	Model	0	1	2–3	4–7	8–15
Position-local, letters	Qwen2.5 7B	0.32	0.38	0.35	0.29	0.26
	Qwen2.5 14B	0.45	0.47	0.41	0.37	0.34
	Qwen2.5 32B	0.60	0.57	0.48	0.45	0.41
	Mistral 7B	0.28	0.34	0.29	0.26	0.25
	Phi-3.5 mini	0.30	0.28	0.28	0.28	0.23
	Qwen3.8 27B	0.68	0.67	0.59	0.51	0.47
	Gemma 4 31B	0.52	0.62	0.55	0.51	0.43
	Gemini 3.5 Flash	0.74	0.69	0.65	0.51	0.46
Position-local, spaced	Qwen2.5 7B	0.30	0.46	0.47	0.43	0.35
	Qwen2.5 14B	0.59	0.67	0.68	0.66	0.59
	Qwen2.5 32B	0.86	0.90	0.89	0.88	0.85
	Mistral 7B	0.31	0.39	0.37	0.27	0.24
	Phi-3.5 mini	0.25	0.37	0.33	0.25	0.21
	Qwen3.8 27B	0.89	0.88	0.91	0.91	0.90
	Gemma 4 31B	0.93	0.94	0.95	0.93	0.94
Position-local, digits	Qwen2.5 7B	0.28	0.38	0.32	0.30	0.25
	Qwen2.5 14B	0.83	0.85	0.86	0.81	0.76
	Qwen2.5 32B	0.90	0.93	0.94	0.93	0.90
	Mistral 7B	0.31	0.38	0.36	0.31	0.26
	Phi-3.5 mini	0.38	0.53	0.52	0.46	0.37
	Qwen3.8 27B	0.90	0.89	0.91	0.91	0.90
	Gemma 4 31B	0.94	0.98	0.99	0.98	0.98
Dependency	Qwen2.5 7B	0.36	0.41	0.42	0.42	0.36
	Qwen2.5 14B	0.45	0.75	0.68	0.60	0.53
	Qwen2.5 32B	0.49	0.85	0.79	0.71	0.63
	Mistral 7B	0.41	0.42	0.45	0.42	0.41
	Phi-3.5 mini	0.40	0.47	0.44	0.40	0.38
	Qwen3.8 27B	0.46	0.70	0.71	0.65	0.58
	Gemma 4 31B	0.49	0.82	0.83	0.81	0.75
	Gemini 3.5 Flash	0.57	1.00	1.00	1.00	0.97

H Marker ablations

I Dose–response robustness

J Well-formedness and cost

Table 6: Reset variants, pooled over positions: TAR Base → Bounds, Δ, and $|z|$, pooled as in Figure 4 (no-marker mean length-matched). Base differs between the original and reset columns because the reset prompt alone shifts no-marker accuracy; Section 6.1 therefore compares marker runs position by position. Originals use 20 to 80 trials per cell (Table 3); reset variants 20, except 80 at Qwen2.5-7B.

		Original (dependency intact)		Reset (dependency severed)	
Task	Model	Base→Bounds	Δ ($\|z\|$)	Base→Bounds	Δ ($\|z\|$)
`running_xor`	7B	0.401→0.403	+0.002 (0.1)	0.450→0.482	+0.032 (3.4)
	32B	0.746→0.798	+0.053 (6.2)	0.715→0.878	+0.163 (11.9)
	Qwen3.8 27B	0.646→0.742	+0.096 (9.5)	0.645→0.855	+0.210 (14.4)
	Gemma 4 31B	0.816→0.850	+0.034 (4.3)	0.816→0.928	+0.112 (10.2)
`running_sum_local`	7B	0.286→0.280	−0.006 (0.8)	0.277→0.348	+0.071 (12.4)
	32B	0.547→0.603	+0.056 (9.0)	0.508→0.660	+0.152 (12.1)
	Qwen3.8 27B	0.441→0.573	+0.131 (13.0)	0.419→0.683	+0.264 (19.4)
	Gemma 4 31B	0.754→0.775	+0.021 (3.2)	0.723→0.840	+0.117 (10.8)
`sliding_window_max`	32B	0.668→0.733	+0.065 (7.9)	0.677→0.816	+0.139 (10.9)
	Qwen3.8 27B	0.666→0.653	−0.013 (1.0)	0.640→0.824	+0.183 (16.4)
	Gemma 4 31B	0.708→0.783	+0.075 (6.3)	0.718→0.817	+0.099 (7.9)
`copy_previous`	7B	0.508→0.418	−0.090 (7.0)	0.565→0.553	−0.013 (1.4)
	32B	0.836→0.818	−0.018 (1.9)	0.874→0.792	−0.083 (4.6)
	Qwen3.8 27B	0.971→0.709	−0.262 (23.9)	0.942→0.836	−0.105 (6.4)
	Gemma 4 31B	0.917→0.832	−0.086 (6.5)	0.837→0.866	+0.029 (1.3)

Table 7: External position-local tasks, pooled across length and density. Base: no-marker accuracy averaged over lengths. Best-k: accuracy at the single best spacing (optimistic). Δ: marker runs minus the length-matched no-marker mean (Section 3). ProcBench-Substitute was regenerated after we made the vendored ProcBench generator's substitution table deterministic (it was drawn in a process-dependent order).

	Qwen2.5-7B			Qwen2.5-32B		
Task	Base	Best-k	Δ ($\|z\|$)	Base	Best-k	Δ ($\|z\|$)
Vowel/consonant tagging	0.438	0.435	−0.083 (2.4)	0.615	0.657	+0.012 (0.5)
ProcBench-Substitute	0.189	0.432	+0.096 (6.7)	0.390	0.757	+0.203 (10.8)
String copying	0.973	1.000	+0.009 (1.0)	0.993	1.000	+0.007 (2.0)

Table 8: Closed-model check and CLRS-Text. Gemini 3.5 Flash with 3 trials per cell, pooled over all tested lengths (including N up to 256 for `cyclic_shift`), with the no-marker mean length-matched. CLRS-Text tasks use token-level markers and scoring.

Task	Model	Base TAR	Bounds TAR	Δ ($\|z\|$)
`cyclic_shift`	Gemini	0.596	0.852	+0.256 (14.4)
`random_permutation`	Gemini	0.561	0.846	+0.284 (8.4)
`conditional_shift`	Gemini	0.453	0.628	+0.175 (3.1)
`running_xor`	Gemini	0.882	0.913	+0.032 (0.4)
`running_sum_local`	Gemini	0.918	0.803	−0.116 (5.2)
`copy_previous`	Gemini	0.984	0.924	−0.060 (2.8)
CLRS minimum	Qwen2.5-7B	0.133	0.183	+0.050 (3.3)
	Qwen2.5-32B	0.592	0.392	−0.200 (4.5)
	Gemini	0.997	0.654	−0.343 (7.3)
CLRS Kadane	Qwen2.5-7B	0.060	0.068	+0.008 (1.0)
	Qwen2.5-32B	0.186	0.122	−0.064 (2.0)
	Gemini	0.384	0.354	−0.030 (0.3)

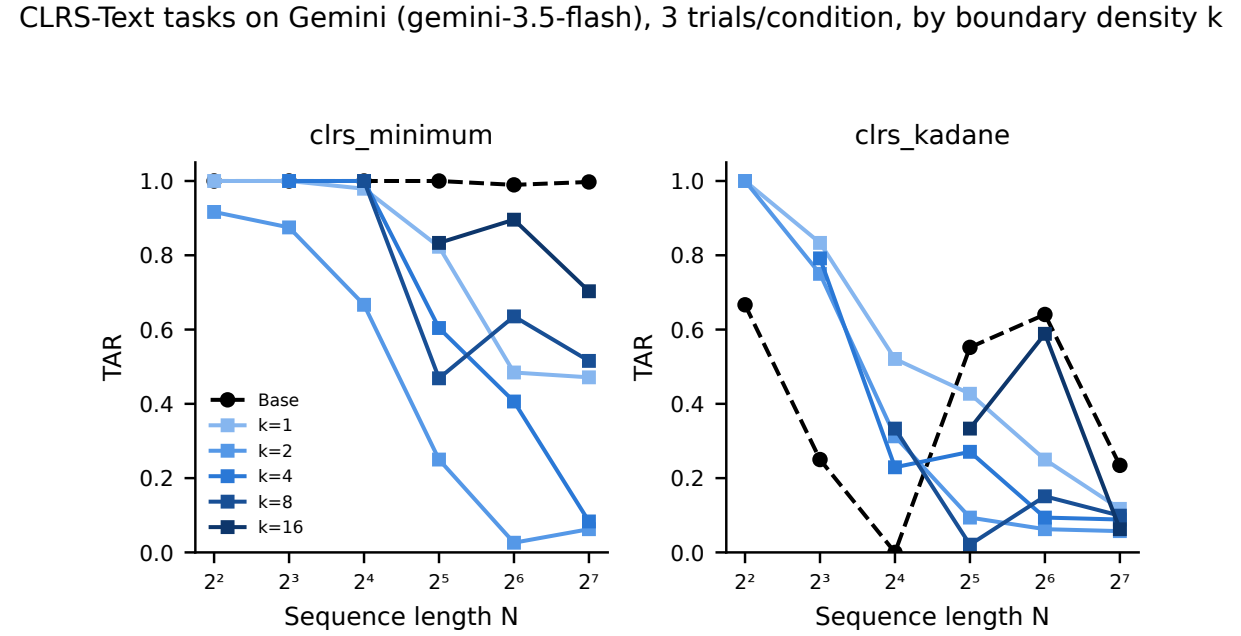


Figure 10: Gemini 3.5 Flash by task and density, with CLRS-Text tasks across models.

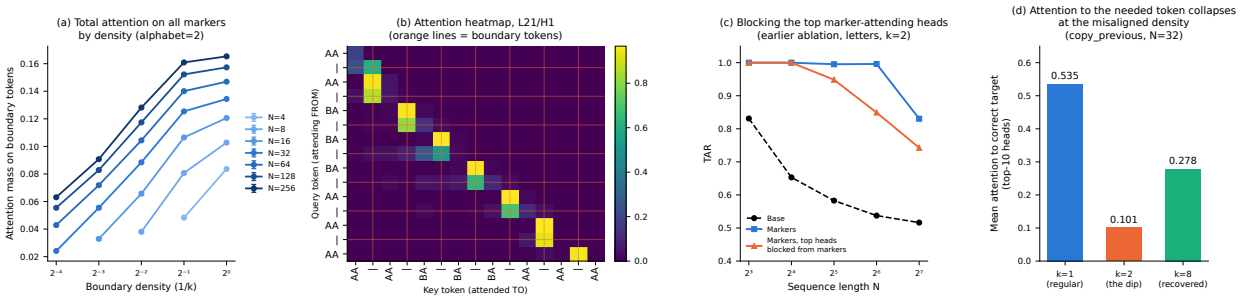


Figure 11: Mechanism (Qwen2.5-32B). (a) Total attention mass on all marker tokens rises with density at every length (`cyclic_shift`, $A = 2$); the number of markers rises faster, so this is not evidence that individual markers draw more attention than content tokens. (b) The most marker-attending head (layer 21, head 1) attends from each position to an adjacent marker. (c) Earlier ablation on contiguous letters ($k = 2$; Section 7): masking the ten most marker-attending heads' attention to markers removes a growing share of the benefit as N grows (0% at $N \leq 16$, 11% at 32, 32% at 64, 28% at 128). (d) `copy_previous` at $N = 32$: mean attention from each output position to the input token it needs (top-10 heads) is 0.535 at $k = 1$, 0.101 at $k = 2$, and 0.278 at $k = 8$, matching accuracy (Base 0.83; $k = 1$: 0.88; $k = 2$: 0.65; $k = 8$: 0.84).

Table 9: Head ablation at Qwen2.5-7B, same design as Table 2. Marker gains are small at this scale, so shares are noisy; "–" marks cells with a gain below 0.03.

				Markers, with heads blocked			Share of gain removed	
Format	N	No markers	Markers	Top 10	Random 10	Top 10, sham	Top 10	Controls
Letters	32	0.17	0.24	0.23	0.24	0.24	14%	0%
Letters	64	0.14	0.16	0.14	0.15	0.16	–	–
Spaced letters	32	0.42	0.55	0.51	0.52	0.55	26%	$\leq$ 22%
Spaced letters	64	0.19	0.30	0.27	0.31	0.32	24%	0%
Digits	32	0.24	0.27	0.28	0.29	0.28	–	–
Digits	64	0.18	0.19	0.19	0.19	0.19	–	–

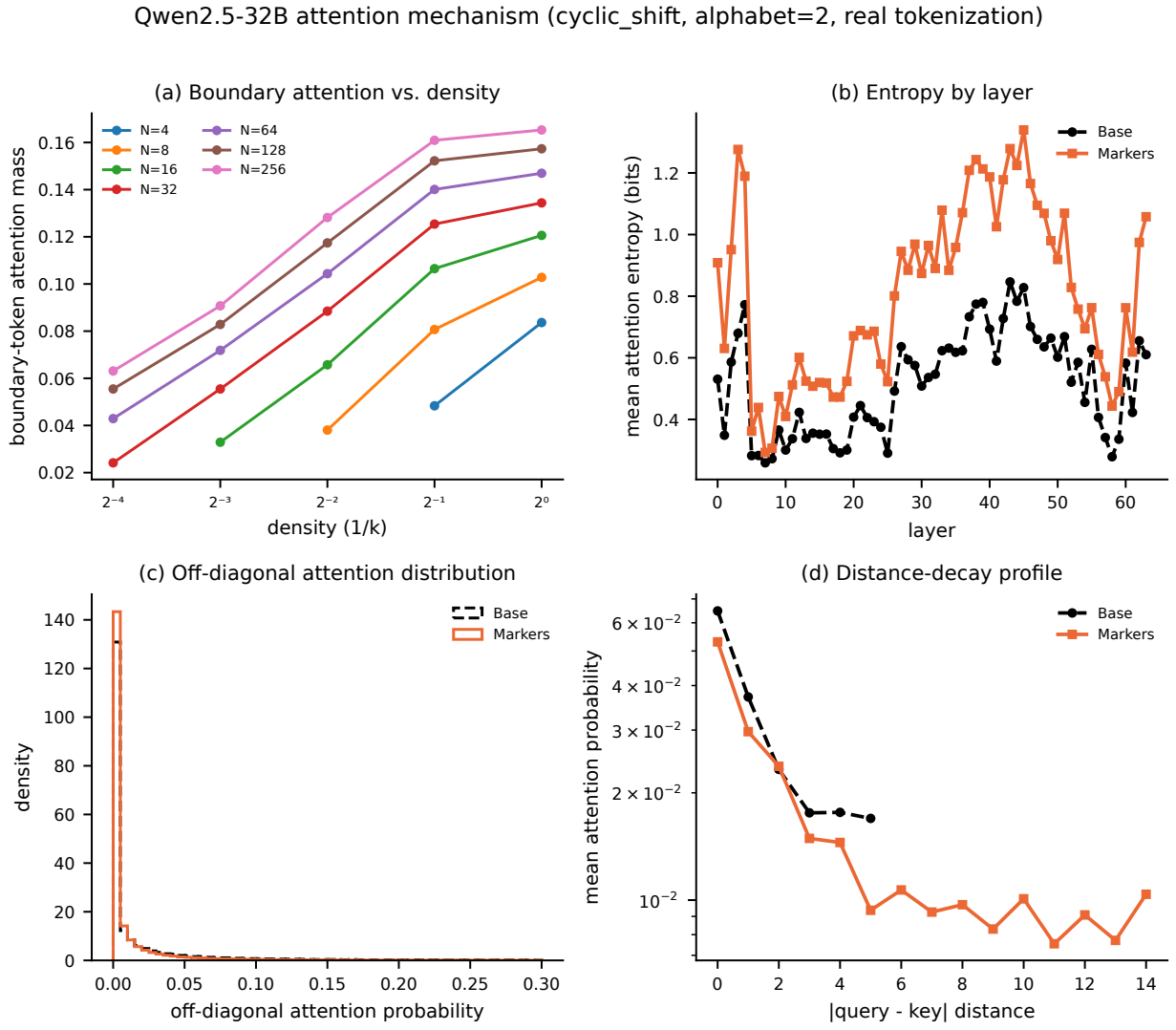


Figure 12: Attention analysis, Qwen2.5-32B, `cyclic_shift`: marker attention by density, attention entropy by layer (higher under markers at nearly every layer), off-diagonal attention, and distance decay.

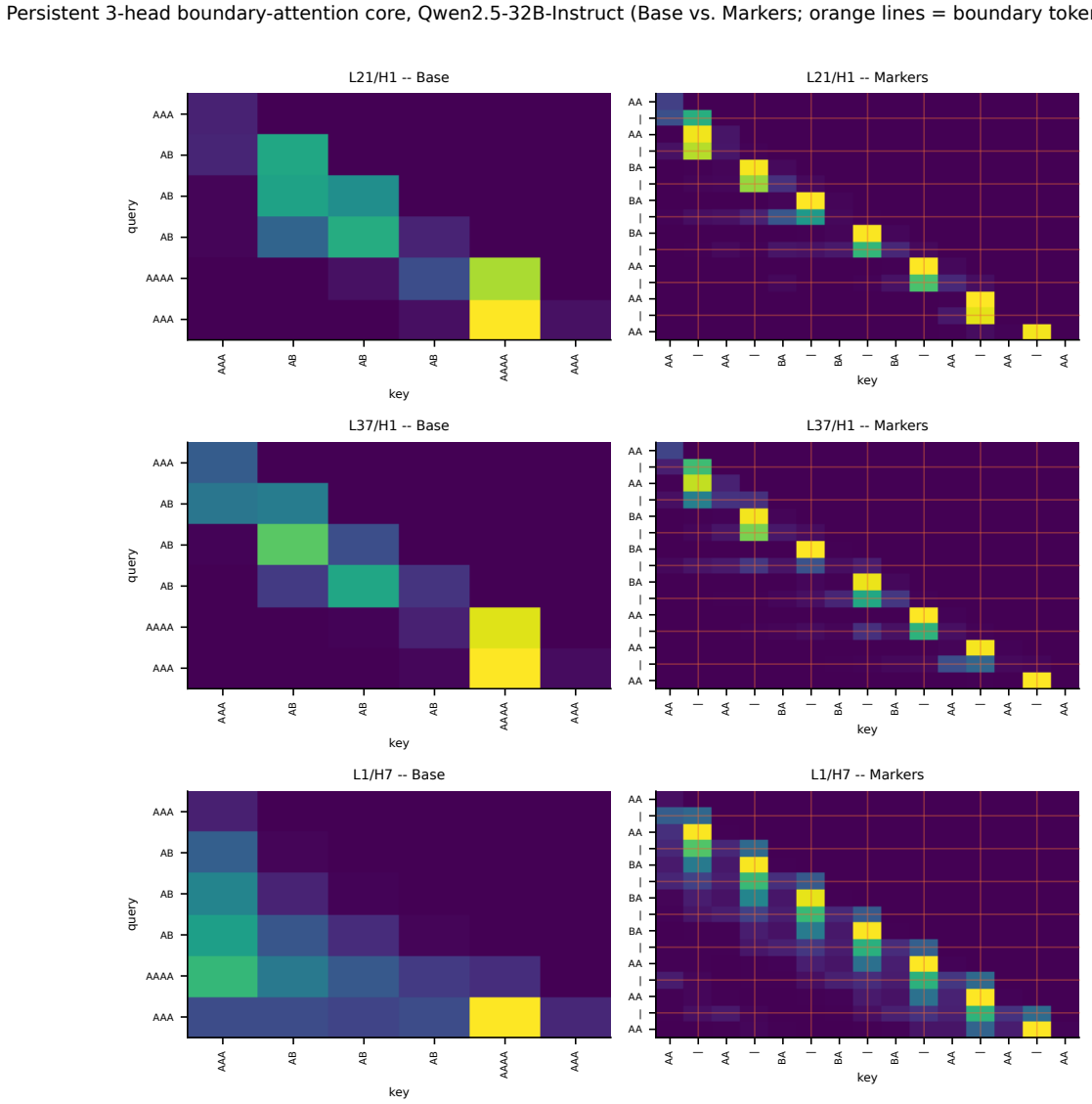


Figure 13: The three most marker-attending heads at Qwen2.5-32B (L21/H1, L37/H1, L1/H7). Without markers, letters merge into multi-character tokens; markers restore per-character resolution.

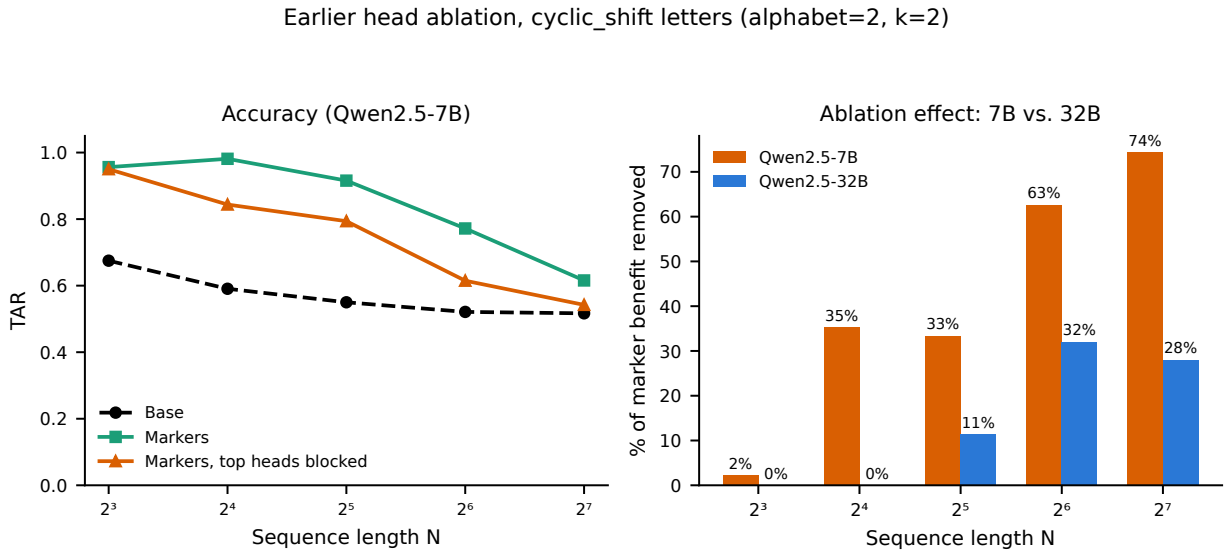


Figure 14: Earlier head ablation, contiguous letters (`cyclic_shift`, $A = 2$, $k = 2$), Qwen2.5-7B and 32B: share of the marker benefit removed by masking the ten most marker-attending heads' attention to markers, re-identified at each length. Unlike Table 2, Base and unablated runs come from an earlier grid and there are no controls at 32B.

Table 10: The ten most marker-attending heads at Qwen2.5-32B, with percentiles within all heads on an induction test and a positional (previous-token) test.

Layer/Head	Marker attention	Induction pct.	Positional pct.
21/1	0.099	38.3	98.3
37/1	0.087	47.4	96.7
1/7	0.086	17.1	97.7
35/19	0.082	23.1	96.5
35/3	0.080	14.7	97.2
30/32	0.079	18.6	96.0
32/37	0.075	14.3	98.8
1/8	0.072	64.8	97.7
32/7	0.071	60.4	90.2
19/19	0.068	9.9	83.9
Mean		30.9	95.3

Table 11: Marker identity, reserved token, and token budget (`cyclic_shift`, $A = 2$, periodic $k = 4$ unless noted). TAR. Budget-matched: `<|>` at $k = 12$ uses the same total tokens as `|` at $k = 4$.

N	Model	Base	`|`	`;`	`#`	Reserved token	`<|>`, $k = 12$
8	7B	0.675	0.717	0.792	0.792	0.469	0.675
	32B	0.831	0.892	0.942	0.908	0.525	0.806
16	7B	0.591	0.825	0.750	0.667	0.216	0.591
	32B	0.653	0.896	0.846	0.912	0.284	0.759
32	7B	0.550	0.679	0.769	0.585	0.092	0.539
	32B	0.583	0.848	0.781	0.744	0.186	0.688
64	7B	0.521	0.616	0.650	0.591	0.049	0.453
	32B	0.537	0.759	0.658	0.693	0.160	0.710
128	7B	0.517	0.545	0.507	0.496	0.046	0.452
	32B	0.516	0.694	0.744	0.561	0.219	0.642

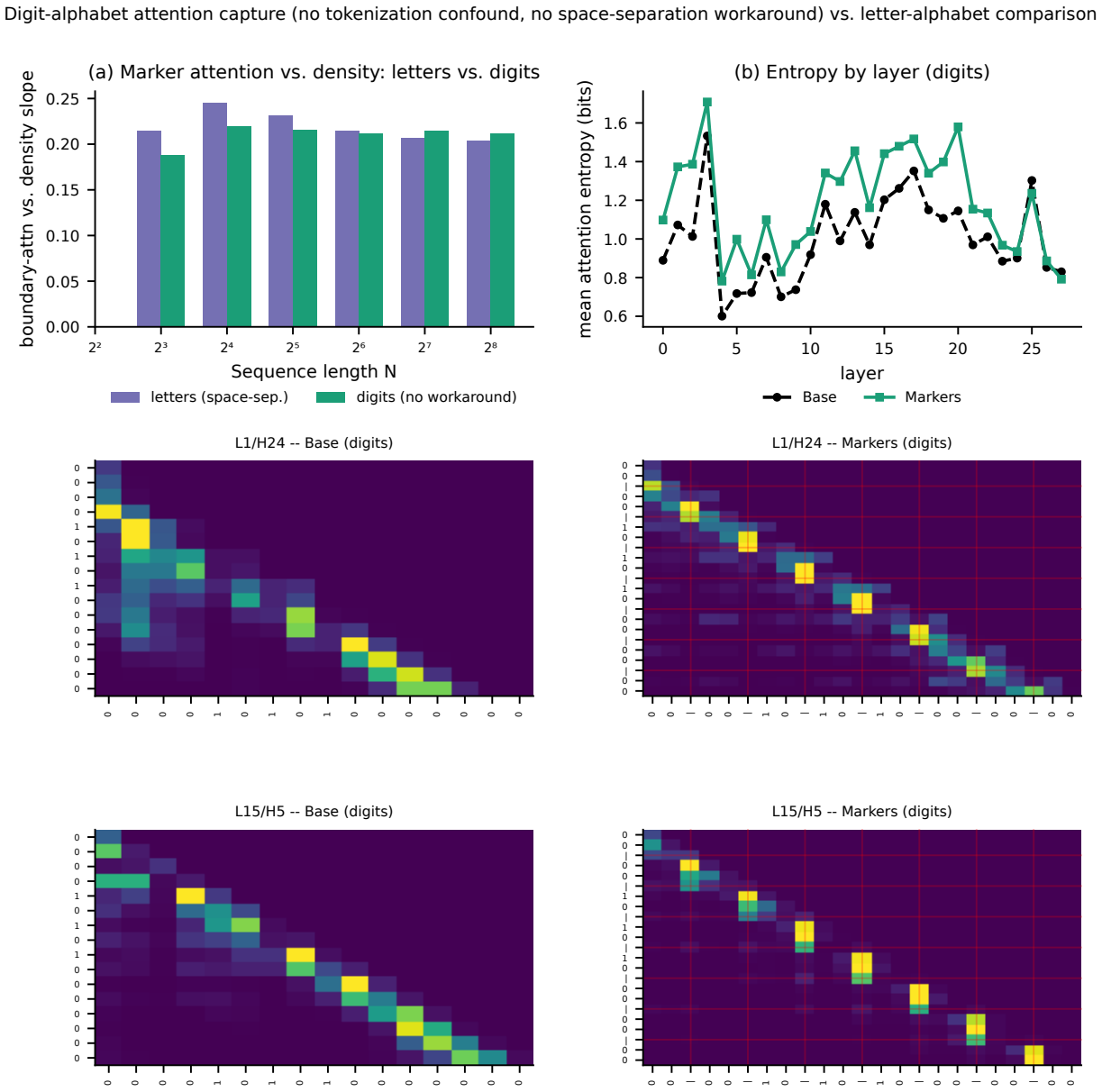


Figure 15: Tokenization check: marker-attention slopes with letters (space-separated, one token per letter) and digits (natively one token per character) are nearly identical.

Table 12: Placement regularity (`cyclic_shift`, $A = 2$, marker count matched to $k = 4$). TAR.

	Qwen2.5-7B					Qwen2.5-32B				
N	Base	Periodic	±1	±2	Random	Base	Periodic	±1	±2	Random
8	0.675	0.717	0.581	0.675	0.633	0.831	0.892	0.869	0.912	0.858
16	0.591	0.825	0.572	0.566	0.583	0.653	0.896	0.863	0.787	0.733
32	0.550	0.679	0.572	0.509	0.552	0.583	0.848	0.847	0.745	0.779
64	0.521	0.616	0.502	0.491	0.491	0.537	0.759	0.731	0.732	0.682
128	0.517	0.545	0.493	0.471	0.452	0.516	0.694	0.691	0.695	0.583

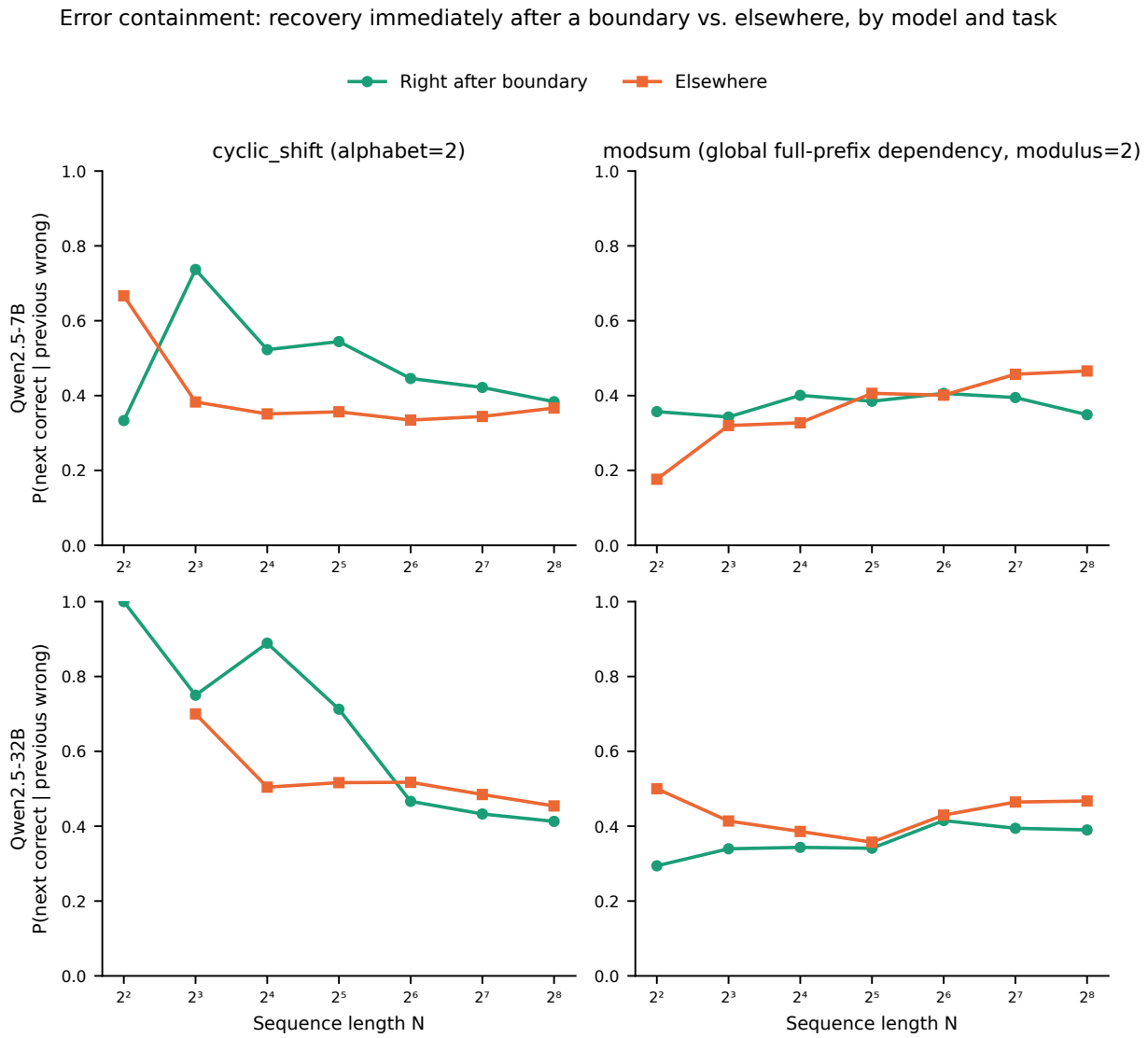


Figure 16: Error recovery: after a wrong output, the fraction of next positions that are correct, split by whether the next position follows a marker. Left: `cyclic_shift`; right: a full-prefix modular sum. $A = M = 2$.

Table 13: Prompt wording (`cyclic_shift`, $A = 2$, $k = 4$, 15 trials per cell). Δ = Bounds − Base TAR.

	Qwen2.5-7B, $N =$					Qwen2.5-32B, $N =$				
Wording	8	16	32	64	128	8	16	32	64	128
Formal	+.09	+.27	+.10	+.13	+.07	+.26	+.17	+.17	+.20	+.18
Casual	−.04	+.08	+.12	+.04	+.10	+.08	+.25	+.17	+.21	+.20
Terse cipher	−.07	+.08	+.00	+.00	+.05	−.12	−.11	+.01	−.07	−.05

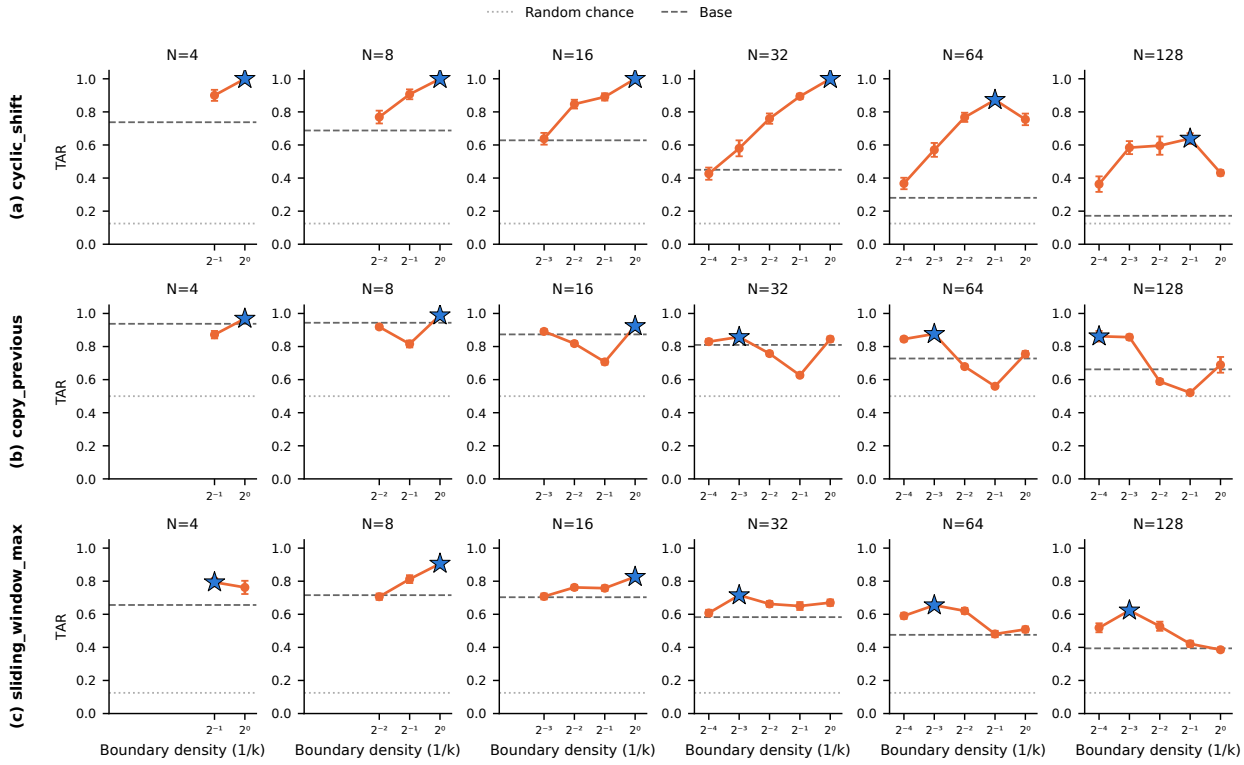


Figure 17: Accuracy by marker density at each length, contiguous letters and dependency tasks (Qwen2.5-32B; $A = 8$ where applicable). Dashed: no-marker baseline; dotted: chance; star: best density.

Table 14: Does a quadratic in density describe accuracy curves? "AIC": share of curves where a quadratic beats linear and exponential fits in-sample. "Held-out": share where it also beats a linear fit on leave-one-density-out prediction.

Task	Tier	Curves	Mean quadratic R^2	AIC	Held-out
`cyclic_shift`	position-local	15	0.876	93%	47%
`random_permutation`	position-local	12	0.941	92%	42%
`conditional_shift`	position-local	12	0.908	100%	50%
`copy_previous`	one-step	4	0.608	50%	0%
`running_xor`	one-step	4	0.449	25%	0%
`running_sum_local`	one-step	12	0.512	42%	0%
`sliding_window_max`	windowed	8	0.473	50%	0%

Table 15: Well-formedness at Qwen2.5-32B. VVR: share of outputs using only the task alphabet. CR: share at least as long as the answer. WFR: share satisfying both. Pooled across lengths and alphabets; first seed batch only, so trial counts are half of Table 3 for the doubly-run tasks.

Task	Condition	Trials	TAR	VVR	CR	WFR
`cyclic_shift`	Base	420	0.511	0.536	0.919	0.521
	Bounds	1740	0.701	0.568	0.806	0.444
`random_permutation`	Base	600	0.499	0.995	0.858	0.855
	Bounds	2280	0.720	0.994	0.923	0.918
`conditional_shift`	Base	300	0.514	0.917	0.937	0.863
	Bounds	1140	0.671	0.834	0.908	0.752
`running_xor`	Base	200	0.781	1.000	0.940	0.940
	Bounds	760	0.796	1.000	0.930	0.930
`running_sum_local`	Base	600	0.590	0.830	0.900	0.765
	Bounds	2280	0.601	0.699	0.946	0.675
`copy_previous`	Base	200	0.859	1.000	0.955	0.955
	Bounds	760	0.817	1.000	0.879	0.879
`sliding_window_max`	Base	200	0.684	1.000	0.995	0.995
	Bounds	760	0.727	1.000	0.943	0.943

Table 16: Wall-clock seconds per call at Qwen2.5-32B. Throughput is 14 to 15 tokens/s in every condition.

Condition	k	$N = 8$	16	32	64	128
Base	–	0.330	0.570	1.270	3.763	11.380
Bounds	1	1.133	2.215	4.403	7.684	14.260
Bounds	2	0.544	1.076	2.142	4.213	9.000
Bounds	4	0.380	0.787	1.729	3.298	6.453
Bounds	8	0.312	0.613	1.288	2.479	4.976

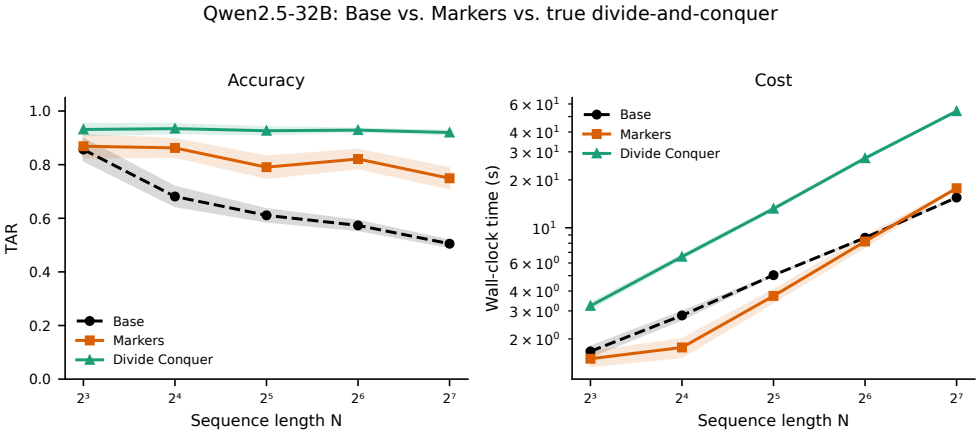


Figure 18: Markers versus true divide-and-conquer (one call per segment), Qwen2.5-32B: accuracy and wall-clock by length.